Hints: Trigonometry – 3. The substitution $x = 2\cos\theta$ and Chebyshev polynomials

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $T_m(T_n(\cos\theta)) = T_m(\cos n\theta) = \cos mn\theta = T_{mn}(\cos\theta)$; two polynomials agreeing on $[-1,1]$ are equal.

Hint 2. A conceptual reason: with $D_n(x) = 2T_n(x/2)$ one has $D_n(z+z^{-1}) = z^n + z^{-n}$, i.e. $D_n \circ h = h \circ z^n$ for $h(z) = z + z^{-1}$. So the Chebyshev family is the "shadow" of the power family z^n, and both are closed under composition ($z^m \circ z^n = z^{mn}$).

Hint 3. Ritt's theorem (to be stated, not proved): if two polynomials of degree > 1 commute and are not iterates of a common polynomial, then after a simultaneous affine change of variable they are both powers x^n or both $\pm T_n$. Check that conjugating T_n by $x \mapsto 2x$ gives D_n, so these are the same family up to affine conjugacy.

Problem 2

Hint 1. Substitute $x = \cos\theta$: $\frac{dx}{\sqrt{1-x^2}}$ becomes $d\theta$ on $[0,\pi]$ and the integral becomes $\int_0^\pi \cos m\theta \cos n\theta\, d\theta$, which is 0 for $m \neq n$ (product-to-sum).

Hint 2. For the expansion write $\cos^n\theta = 2^{-n}(e^{i\theta} + e^{-i\theta})^n$ and pair the terms $e^{\pm i(n-2k)\theta}$: $x^n = 2^{1-n}\sum_{0\leqslant k<n/2}\binom{n}{k}T_{n-2k}(x)$, plus $2^{-n}\binom{n}{n/2}$ when n is even.

Problem 3

Hint 1. $U_{n-1}(\cos\theta) = \frac{\sin n\theta}{\sin\theta}$ vanishes at $\theta = \frac{k\pi}{n}$, $k = 1,\dots,n-1$; these give $n-1$ distinct roots of a polynomial of degree $n-1$.

Hint 2. Let J_n be the $n \times n$ tridiagonal matrix. Expanding $P_n(x) = \det(xE - J_n)$ along the last row gives $P_n = xP_{n-1} - P_{n-2}$, $P_0 = 1$, $P_1 = x$. Compare with the recurrence for $U_n(x/2)$: $P_n(x) = U_n(x/2)$, whose roots are $2\cos\frac{k\pi}{n+1}$, $k = 1,\dots,n$.

Hint 3. Wirtinger: for $x_0 = x_n = 0$, $\sum_{k=1}^n (x_k - x_{k-1})^2 = x^{\mathsf{T}}(2E - J_{n-1})x$ with $x = (x_1,\dots,x_{n-1})$. Its smallest eigenvalue is $2 - 2\cos\frac{\pi}{n} = 4\sin^2\frac{\pi}{2n}$, which is the sharp constant, attained by the eigenvector $x_k = \sin\frac{k\pi}{n}$.

Problem 4

Hint 1. Use the affine map $x = \frac{a+b}{2} + \frac{b-a}{2}t$, $t \in [-1,1]$. A monic polynomial of degree n in x becomes $\left(\frac{b-a}{2}\right)^n$ times a monic polynomial in t.

Hint 2. By Chebyshev's theorem the minimum is $\left(\frac{b-a}{2}\right)^n 2^{1-n} = 2\left(\frac{b-a}{4}\right)^n$, attained only by $p(x) = 2\left(\frac{b-a}{4}\right)^n T_n\left(\frac{2x-a-b}{b-a}\right)$.

Problem 5

Hint 1. $\cos 7\theta = -1$ for $\theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \pi$. So $\cos\frac{\pi}{7}, \cos\frac{3\pi}{7}, \cos\frac{5\pi}{7}$ are roots of $T_7(x) + 1 = 64x^7 - 112x^5 + 56x^3 - 7x + 1$, which is divisible by $x + 1$.

Hint 2. One checks $T_7(x)+1=(x+1)(8x^3-4x^2-4x+1)^2$, so the three distinct numbers above are exactly the roots of $x^3-\frac{1}{2}x^2-\frac{1}{2}x+\frac{1}{8}$. Alternatively, find this cubic from Vieta: compute $\sum\cos\frac{(2j+1)\pi}{7}$, the sum of pairwise products and the product with product-to-sum formulas.

Hint 3. Irreducibility: by the rational root theorem a rational root of $8x^3-4x^2-4x+1$ is among $\pm1,\pm\frac{1}{2},\pm\frac{1}{4},\pm\frac{1}{8}$; none works, and a cubic without rational roots is irreducible. Since $\cos\frac{2\pi}{7}=-\cos\frac{5\pi}{7}$, its minimal polynomial is obtained by $x\mapsto -x$; $\cos\frac{3\pi}{7}$ is another root of the first cubic.

Problem 6

Hint 1. With $D_n(x)=2T_n(x/2)$ the recurrence $D_{n+1}=xD_n-D_{n-1}$, $D_0=2$, $D_1=x$, shows that D_n is monic with integer coefficients, and $D_n(2\cos\frac{2\pi}{n})=2\cos 2\pi=2$.

Hint 2. So $2\cos\frac{2\pi}{n}$ is a root of the monic integer polynomial $D_n(x)-2$; by the rational root theorem, if it is rational it is an integer in $[-2,2]$. Check which n give $2,-2,0,\pm1$: $n=1,2,4,3,6$.

Problem 7

Hint 1. Suppose $|p(x_0)|>|T_n(x_0)|$ for some $x_0>1$ (the case $x_0<-1$ is symmetric). Put $\lambda=T_n(x_0)/p(x_0)$, so $|\lambda|<1$, and consider $q=T_n-\lambda p$.

Hint 2. $q(x_0)=0$, and at the $n+1$ points $\cos\frac{k\pi}{n}$ the sign of q is that of $T_n=(-1)^k$, because $|\lambda p|<1$. So q has $n+1$ zeros and degree $\leqslant n$: $q\equiv 0$, i.e. $T_n=\lambda p$, which contradicts $T_n(1)=1>|\lambda|$.

Hint 3. For the leading coefficient c: $|c|=\lim_{x\to\infty}|p(x)|/x^n\leqslant\lim_{x\to\infty}|T_n(x)|/x^n=2^{n-1}$.

Problem 8

Hint 1. Multiply $G(t)=\sum_{k\geqslant 0}T_k(x)t^k$ by $1-2xt+t^2$ and use the recurrence: all coefficients of t^k, $k\geqslant 2$, vanish, leaving $T_0+(T_1-2xT_0)t=1-xt$. Hence $G(t)=\frac{1-tx}{1-2tx+t^2}$, and the same computation gives $\sum_k U_kt^k=\frac{1}{1-2tx+t^2}$.

Hint 2. The addition formula is $\cos(m+n)\theta+\cos(m-n)\theta=2\cos m\theta\cos n\theta$ written with $x=\cos\theta$ (for $m\geqslant n$); a polynomial identity valid on $[-1,1]$ holds everywhere. (With generating functions: $\frac{1-tx}{1-2tx+t^2}=\frac{1}{2}\big(\frac{1}{1-tz}+\frac{1}{1-t/z}\big)$ for $x=\frac{z+z^{-1}}{2}$, i.e. $T_k=\frac{z^k+z^{-k}}{2}$, and the formula becomes an identity for powers of z.)

Problem 9

Hint 1. For $x=2\cos\theta$: $x^3-3x=2\cos 3\theta$. If $|c|\leqslant 2$, put $\cos 3\theta=\frac{c}{2}$; the three real roots are $2\cos(\theta_0+\frac{2\pi k}{3})$, $k=0,1,2$, with $\theta_0=\frac{1}{3}\arccos\frac{c}{2}$.

Hint 2. If $c>2$, $x=2\cosh t$ gives $2\cosh 3t=c$, so the unique real root is $2\cosh\big(\frac{1}{3}\operatorname{arccosh}\frac{c}{2}\big)=u+u^{-1}$ with $u^3=\frac{c+\sqrt{c^2-4}}{2}$ – exactly Cardano's formula; for $c<-2$ use $x=-2\cosh t$.

Hint 3. For $|c|<2$ Cardano gives $u^3=\frac{c\pm i\sqrt{4-c^2}}{2}=e^{\pm 3i\theta_0}$: the roots are $u+\bar{u}=2\cos\theta_0$ with u a cube root of a non-real number. That is the casus irreducibilis: the real roots require trisecting an angle, i.e. extracting cube roots of complex numbers.

Problem 10

Hint 1. The condition says the value at 0 dominates the ℓ^2-mass on the integer points $-A,\dots,-1,1,\dots,B$. Look for a polynomial that equals 1 at 0 and decays like $1/|j|$ at the point j, with a small constant: then $\sum_{j\neq 0}p(j)^2\leqslant\text{const}^2\cdot\frac{\pi^2}{3}<1$.

Hint 2. Lagrange basis polynomials at Chebyshev nodes do this. If $y_0 = \cos\theta_0$ is a zero of T_m, then $\ell(y) = \frac{T_m(y)}{(y-y_0)T_m'(y_0)}$ has degree $m-1$, $\ell(y_0) = 1$, and $|\ell(y)| \leqslant \frac{\sin\theta_0}{m|y-y_0|}$ on $[-1,1]$, since $|T_m'(y_0)| = \frac{m}{\sin\theta_0}$.

Hint 3. Use an affine map $y = y_0 + x/R$ sending $[-A, B]$ into $[-1,1]$ and 0 to the node y_0. Then $|\ell(y_0 + j/R)| \leqslant \frac{R\sin\theta_0}{m|j|}$, so you need $R\sin\theta_0 \leqslant cm$ with a small constant c, for $m \approx C\sqrt{AB}$.

Hint 4. Let $A \leqslant B$. The interval of length $A+B$ must fit: $R \geqslant \max(\frac{A}{1+y_0}, \frac{B}{1-y_0})$. Choose the node with $\varphi = \pi - \theta_0$ of order $\sqrt{A/B}$ (Chebyshev nodes are $\frac{\pi}{m}$ apart in angle, and $\frac{\pi}{m} \lesssim \sqrt{A/B}$ once C is large): then $1 + y_0 \approx \frac{\varphi^2}{2}$, $R \approx B$ and $R\sin\theta_0 \approx B\varphi \approx \sqrt{AB}$. With $m = \lceil C\sqrt{AB}\rceil$ and C large the ratio $\frac{R\sin\theta_0}{m}$ is small. (When A and B are comparable a node near $\theta_0 = \frac{\pi}{2}$ and $R \approx A + B$ do the job.)